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Bachelor thesis

Symbolic Reasoning for LLM-Based Service Composition

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Description

This thesis looks at how automated service composition in Service-Oriented Computing can be improved by combining large language models with symbolic reasoning. The goal is to increase generation accuracy and to produce more accurate service compositions from natural-language tasks and service descriptions.

Problem Statement

Automated service composition is an important problem in Service-Oriented Computing, but classical methods are difficult to use in practice because they require a lot of formal modeling effort [1]. LLM-based approaches like Compositio Prompto make this task easier by combining natural-language tasks with OpenAPI specifications and generating code for a service composition [2]. Still, the results are not fully accurate and can fail in sequence selection, constraint handling, and output validity.

Recent prior work introduced an initial framework and prototype for integrating symbolic reasoning into LLM-based service composition [3]. In this approach, generated outputs are analyzed with external tools such as static code analysis and linters, and the resulting feedback is fed back into the generation loop, similar in spirit to Reflexion and CRITIC [4, 5]. Yet important questions remain open: how this feedback should be represented, how it should influence subsequent generation steps, and whether it yields measurable improvements over an LLM-only baseline. This thesis addresses these gaps through a systematic follow-up investigation in the SOC domain.

Research Question

The guiding research question of this thesis is:

> How can symbolic reasoning be integrated into LLM-based automated service composition to improve generation accuracy in Service-Oriented Computing?

To make this question more concrete, the thesis will also look at the following sub-questions: What kind of feedback from analysis tools is most useful? How should symbolic feedback be

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added to the generation loop? And how much better can the improved approach perform compared to a baseline LLM-only composition workflow?

Methodology

The thesis will follow an iterative design-and-evaluate approach:

1. Review the relevant literature on automated service composition, LLM-based composition, and tool-assisted symbolic reasoning to derive a concrete system design.
2. Implement a prototype that uses service descriptions, task prompts, and schema information as input and produces candidate compositions as output.
3. Extend the prototype with a symbolic reasoning layer that validates, critiques, or repairs generated outputs by using concrete external analysis tools, such as Spectral for OpenAPI linting, schema validators for input-output checks, Ruff/mypy for generated Python code, and SOCBench-SC for endpoint-level static analysis.
4. Define the evaluation protocol by separating structural from functional feedback: static code analysis and linting detect syntactic, schema, and consistency issues, while functional correctness is assessed against benchmark references.
5. Evaluate both the baseline LLM-only pipeline and the enhanced pipeline on benchmarks such as SOCBench-D and RestBench, using their tasks and associated service descriptions.
6. Measure precision, recall, and the rate of valid service compositions to assess whether analysis-based feedback improves generation quality and reliability.
7. Discuss the results qualitatively and quantitatively to identify the strengths and weaknesses of the proposed method.

Concepts and implementation produced in this project must be compatible and published under the MIT license.

Keywords: automated service composition, large language models, symbolic reasoning, analysis tools, service-oriented computing

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